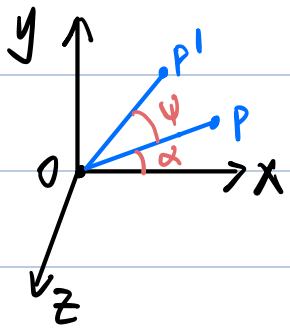
① 绕固定轴 Z 轴旋转



$$P(x,y)=(\cos\alpha,\sin\alpha)$$

$$P'(x',y')=(\cos(\alpha+\psi),\sin(\alpha+\psi))$$

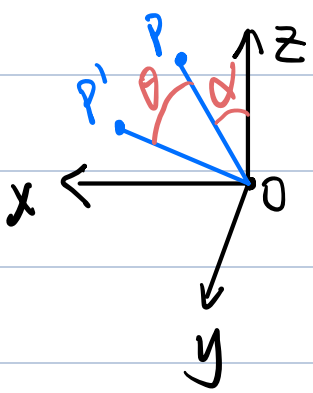
$$x'=\cos\alpha\cos\psi-\sin\alpha\sin\psi=x\cos\psi-y\sin\psi$$

$$y'=\sin\alpha\cos\psi+\cos\alpha\sin\psi=y\cos\psi+x\sin\psi$$

$$z'=z$$

$$R_z(\psi)=\begin{bmatrix}\cos\psi & -\sin\psi & 0\\ \sin\psi & \cos\psi & 0\\ 0 & 0 & 1\end{bmatrix}$$

② 绕固定轴 Y 轴旋转



$$P(x,z)=(\sin\alpha,\cos\alpha)$$

$$P'(x',z')=(\sin(\alpha+\theta),\cos(\alpha+\theta))$$

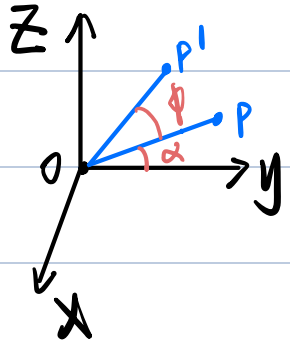
$$x'=\sin\alpha\cos\theta+\cos\alpha\sin\theta=x\cos\theta+z\sin\theta$$

$$y'=y$$

$$z'=\cos\alpha\cos\theta-\sin\alpha\sin\theta=z\cos\theta-x\sin\theta$$

$$R_y(\theta)=\begin{bmatrix}\cos\theta & 0 & \sin\theta\\ 0 & 1 & 0\\ -\sin\theta & 0 & \cos\theta\end{bmatrix}$$

③ 绕固定轴X轴旋转



根据绕Z轴旋转的例子：
$\begin{cases} x \Rightarrow y \\ y \Rightarrow z \\ z \Rightarrow x \end{cases}$

$P(x,y)=(\cos\alpha,\sin\alpha)$

$P'(x',y')=(\cos(\alpha+\psi),\sin(\alpha+\psi))$

$x'=\cos\alpha\cos\psi-\sin\alpha\sin\psi=x\cos\psi-y\sin\psi$

$y'=\sin\alpha\cos\psi+\cos\alpha\sin\psi=y\cos\psi+x\sin\psi$

$z'=z$

$x'=x$

$y'=y\cos\phi-z\sin\phi$

$z'=z\cos\phi+y\sin\phi$

$$R_x(\phi)=\begin{bmatrix}1 & 0 & 0\\ 0 & \cos\phi & -\sin\phi\\ 0 & \sin\phi & \cos\phi\end{bmatrix}$$

④ 刚体旋转的姿态矩阵R

$$\begin{pmatrix}x'\\ y'\\ z'\end{pmatrix}=R\begin{pmatrix}x\\ y\\ z\end{pmatrix}=R_xR_yR_z\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$$R=R_xR_yR_z=\begin{bmatrix}1 & 0 & 0\\ 0 & \cos\phi & -\sin\phi\\ 0 & \sin\phi & \cos\phi\end{bmatrix}\begin{bmatrix}\cos\theta & 0 & \sin\theta\\ 0 & 1 & 0\\ -\sin\theta & 0 & \cos\theta\end{bmatrix}\begin{bmatrix}\cos\psi & -\sin\psi & 0\\ \sin\psi & \cos\psi & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}\cos\theta & 0 & \sin\theta\\ \sin\phi\sin\theta & \cos\phi & -\sin\phi\cos\theta\\ -\sin\theta\cos\phi & \sin\phi & \cos\phi\cos\theta\end{bmatrix}\begin{bmatrix}\cos\psi & -\sin\psi & 0\\ \sin\psi & \cos\psi & 0\\ 0 & 0 & 1\end{bmatrix}$$

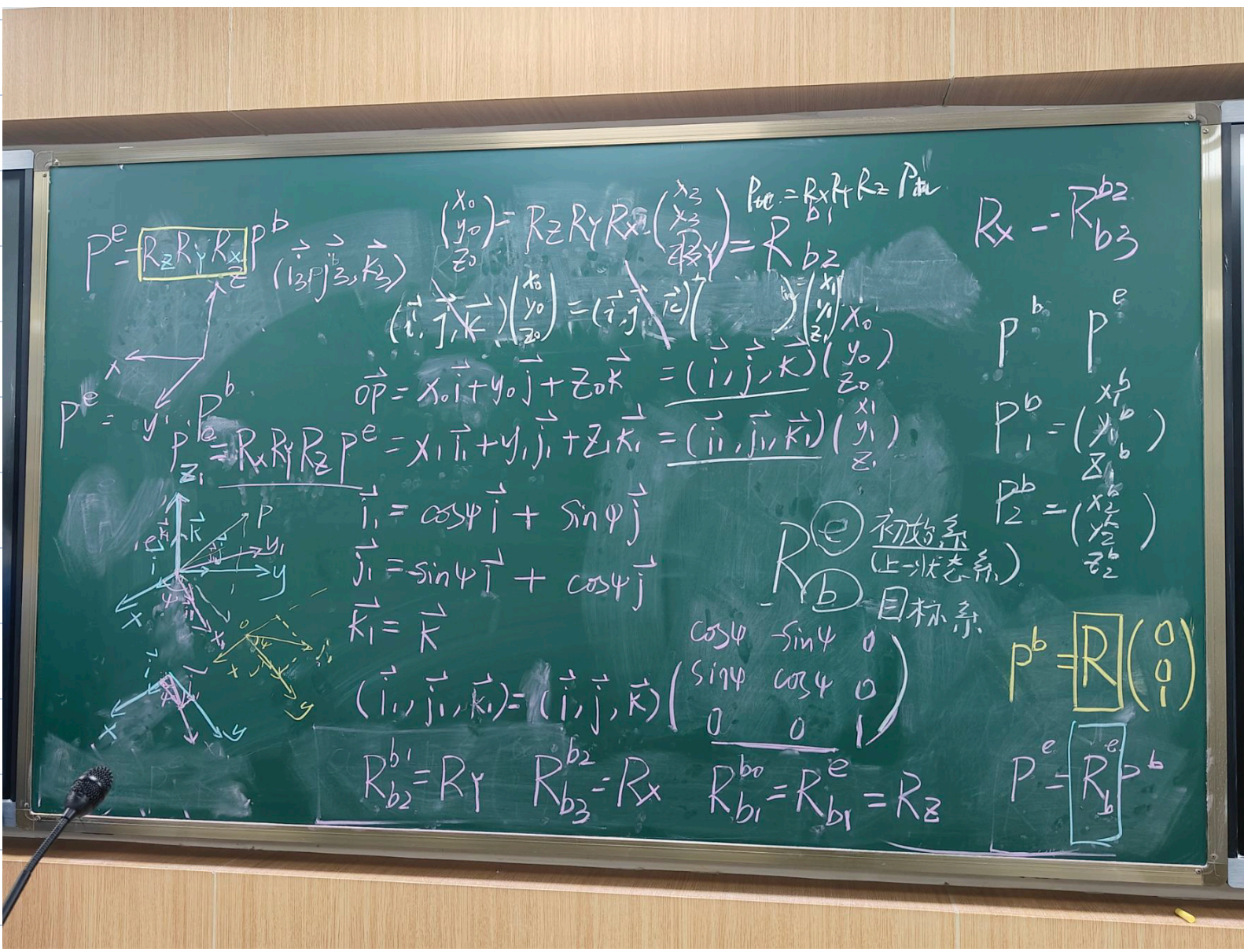
$$= \begin{bmatrix} \cos\theta\cos\psi & -\cos\theta\sin\psi & \sin\theta \\ \sin\phi\sin\theta\cos\psi + \cos\phi\sin\psi & -\sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & -\sin\phi\cos\theta \\ -\sin\theta\cos\phi\cos\psi + \sin\phi\sin\psi & \sin\theta\cos\phi\sin\psi + \sin\phi\cos\psi & \cos\phi\cos\theta \end{bmatrix}$$

- $P'^{e} = R_x R_y R_z P^{e}$

用途：已知刚体向量在地系的初始坐标为 P^e

左乘 $R_x R_y R_z$ 后.

⇒ 刚体向量在地系下外旋后的新坐标 P'^e.



$$\vec{OP} = x_0\vec{i} + y_0\vec{j} + z_0\vec{k} = (\vec{i}, \vec{j}, \vec{k})\begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix}$$
$$= x_1\vec{i}_1 + y_1\vec{j}_1 + z_1\vec{k}_1 = (\vec{i}_1, \vec{j}_1, \vec{k}_1)\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$$
$$= x_2\vec{i}_2 + y_2\vec{j}_2 + z_2\vec{k}_2 = (\vec{i}_2, \vec{j}_2, \vec{k}_2)\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$$
$$= x_3\vec{i}_3 + y_3\vec{j}_3 + z_3\vec{k}_3 = (\vec{i}_3, \vec{j}_3, \vec{k}_3)\begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix}$$

$$R_{b_1}^{e} = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} = R_z(\psi)$$ 绕机体z轴

$$R_{b_2}^{b_1} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} = R_y(\theta)$$ 绕机体y轴

$$R_{b_3}^{b_2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix} = R_x(\phi)$$ 绕机体x轴.

$(\vec{i}_1, \vec{j}_1, \vec{k}_1) = (\vec{i}, \vec{j}, \vec{k})R_{b_1}^{e}$ 代入 $(\vec{i}_1, \vec{j}_1, \vec{k}_1)\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = (\vec{i}, \vec{j}, \vec{k})\begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix}$.

$$\Rightarrow (\vec{i}, \vec{j}, \vec{k})\begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} = (\vec{i}, \vec{j}, \vec{k})R_{b_1}^{e}\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \Rightarrow \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} = R_{b_1}^{e}\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$$ ①

同理可得: $\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = R_{b_2}^{b_1}\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$ ② $\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = R_{b_3}^{b_2}\begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix}$ ③

由①②③⇒

$$\begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} = R_{b_1}^{e}R_{b_2}^{b_1}R_{b_3}^{b_2}\begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix} = R_{b_3}^{e}\begin{pmatrix} x_3 \\ y_3 \\ z \end{pmatrix}$$

$= R_z R_y R_x \begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix}$

$\Rightarrow \quad P^{bo} = R_z R_y R_x P^{b3}$ 又：$P^{bo} = P^e$

P^{b3} 可以理解为目标机系b下的机坐标系.

$\Rightarrow$ $\boxed{P^e = R_z R_y R_x P^b}$

$\Rightarrow$ $\boxed{P^e = R_b^e P^b}$

$R_z R_y R_x$:

$$\begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}$$

$$= \begin{bmatrix} \cos\psi\cos\theta & -\sin\psi & \cos\psi\sin\theta \\ \sin\psi\cos\theta & \cos\psi & \sin\psi\sin\theta \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}$$

$$= \begin{bmatrix} -\cos\psi\cos\theta & -\sin\psi\cos\phi + \cos\psi\sin\theta\sin\phi & \sin\psi\sin\phi + \cos\psi\sin\theta\cos\phi \\ \sin\psi\cos\theta & \cos\psi\cos\phi + \sin\psi\sin\theta\sin\phi & -\cos\psi\sin\phi + \sin\psi\sin\theta\cos\phi \\ -\sin\theta & \cos\theta\sin\phi & \cos\theta\cos\phi \end{bmatrix}$$

令 $\theta=\pm\frac{\pi}{2}$，以 $\theta=\frac{\pi}{2}$ 为例 $\sin\theta=1$ $\cos\theta=0$

$$=\begin{bmatrix} 0 & -\sin\psi\cos\phi+\cos\psi\sin\phi & \sin\psi\sin\phi+\cos\psi\cos\phi \\ 0 & \cos\psi\cos\phi+\sin\psi\sin\phi & -\cos\psi\sin\phi+\sin\psi\cos\phi \\ -1 & 0 & 0 \end{bmatrix}$$

$$=\begin{bmatrix} 0 & -\sin(\psi-\phi) & \cos(\psi-\phi) \\ 0 & \cos(\psi-\phi) & \sin(\psi-\phi) \\ -1 & 0 & 0 \end{bmatrix}$$

姿态矩阵 R 取决于 θ 和 $(\psi-\phi)$，即失去一个自由度，只有两个自由度

$\Rightarrow$ 万向锁

证明旋转矩阵为正交矩阵

$$R_z=\begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad R_z^T=\begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_z\cdot R_z^T=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}=I \Rightarrow R_z\text{为正交矩阵}$$

同理可得：R_y、R_x 也为正交矩阵

又：所有的旋转矩阵都可以表示为 R_x、R_y、R_z 的任意乘积

∴ 旋转矩阵为正交矩阵 $\begin{cases} ① R^T = R^{-1} \\ ② |R| = \pm 1 \\ ③ \text{任意两个正交矩阵的乘积为正交矩阵} \end{cases}$

eg1: $R_e^b = (R_b^e)^{-1} = R_b^{eT}$

eg2: $P^e = R_z R_y R_x P^b$

$(R_z R_y R_x)^{-1} P^e = P^b$

$(R_z R_y R_x)^T P^e = P^b$

$R_x^T R_y^T R_z^T P^e = P^b$

综上所述: ① $P^b = R_x R_y R_z P^b$

② $P^e = R_b^e P^b$

③ $P^e = R_z R_y R_x P^b$

④ $R_z^T R_y^T R_x^T P^e = P^b$

⑤ R_z、R_y、R_x 及相乘组合为正交矩阵

$R_z^T = R_z^{-1}$

⑥ $R_{b2}^{b1} = (R_{b1}^{b2})^T = (R_{b1}^{b2})^{-1}$